SAGE-HE

Secure Adaptive Gated Ensemble for

Privacy-Preserving Steel Surface Defect Classification

RESEARCH, ARCHITECTURE & IMPLEMENTATION

SUHAIB S Z

NEU-CLS · EfficientNet-B0 + multi-scale local branch · CKKS homomorphic inference

Abstract

— **Problem.** Steel surface inspection is increasingly outsourced to cloud GPUs, but production imagery is commercially sensitive and leaks line yield and defect rates.

— **Base paper.** Jia et al. (2023) encrypt Sobel-selected image chunks under CKKS and report 97.1% on NEU-CLS — without precision, recall, split sizes or epoch count.

— **Proposal.** SAGE-HE pairs an EfficientNet-B0 global branch with a multi-scale local branch, selects chunks by a learned Sobel-conditioned attention rather than a fixed threshold, fuses the branches with a channel-wise gate, and compresses to a 64-d vector that is encrypted and classified by a depth-3 polynomial head.

— **Result.** SAGE-HE reaches 100.00% test accuracy against 93.98% for our faithful reproduction of the baseline, over 3 seeds on an identical held-out split. Encrypted and plaintext inference agree on 100.0% of test images.

— **Caveat, and the main finding.** NEU-CLS is saturated, so component contributions are measured in a reduced-data regime. There the full model scores *below* the bare EfficientNet-B0 backbone on every seed, and the gap closes to nothing once data is less scarce. The architectural claim is withdrawn; the reproduction, the encrypted inference and the evaluation protocol stand.

Introduction

Why surface defect classification, and why privacy changes the problem

— Hot-rolled steel strip is inspected continuously; surface defects determine whether a coil is sold, downgraded or scrapped.

— Modern classifiers are accurate but compute-hungry, which pushes inference to cloud GPUs.

— Inspection images are a direct readout of process health: defect mix and rate expose yield, machine condition and production volume.

— A mill that sends raw images to a third-party cloud is disclosing exactly the data it least wants a competitor to hold.

The tension

Accuracy wants a large model on someone else's hardware.

Confidentiality wants the data never to leave the plant.

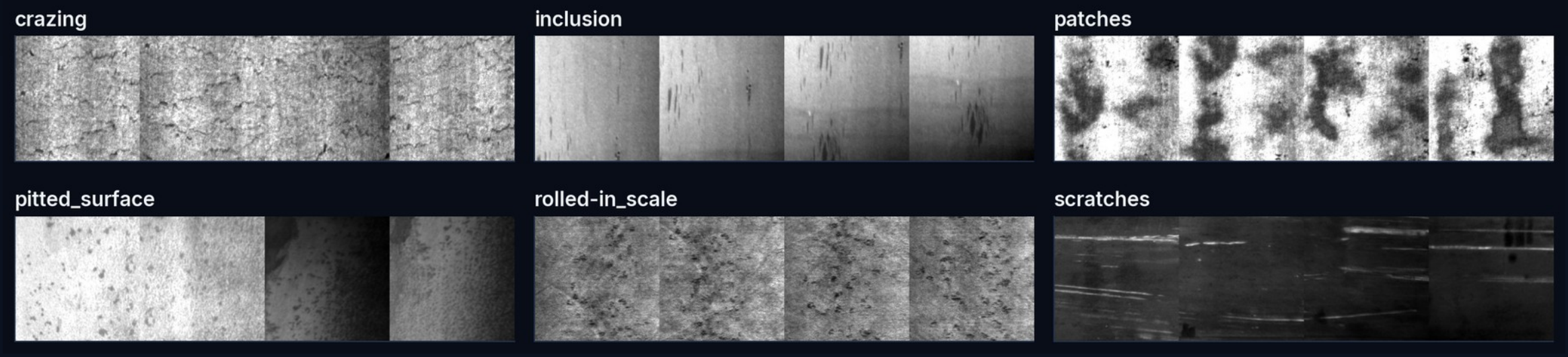
Homomorphic encryption is the one mechanism that can serve both.

What this project does

Builds a working system on that boundary, measures its real cost, and states precisely what it does and does not protect.

Industrial Problem Context

The six NEU-CLS defect classes



Each class has a different physical cause and a different spatial scale — which is the architectural argument for a multi-scale local branch.

Problem Statement

Can a cloud service classify steel surface defects at state-of-the-art accuracy without ever being able to read the images or the predictions — and what does that cost?

— **Accuracy.** The privacy mechanism must not meaningfully degrade classification.

— **Practicality.** Encrypted inference must run in seconds, not hours, on commodity CPU.

— **Honesty of the threat model.** It must be stated exactly which party learns what — a scheme that protects less than it appears to is worse than none.

— **Reproducibility.** The comparison against prior work must use the same held-out data, not a number quoted from a different split.

Research Motivation

What the base paper leaves open

— **Selection is fixed and blind.** Jia et al. keep chunks where the Sobel gradient exceeds a threshold τ. The rule is non-differentiable, cannot adapt to the task, and scores a lighting seam exactly as highly as a genuine defect. τ is never reported.

— **The architecture is minimal.** One 7×7 convolution and two fully connected layers — no mechanism for defects that live at different spatial scales.

— **Evaluation is thin.** Accuracy only; no precision, recall, F1, confusion matrix, split sizes or epoch count for NEU-CLS.

— **The headline comparison is cross-dataset.** Table 3 ranks their NEU-CLS accuracy against other papers' results on cancer genomics, Fashion-MNIST and CIFAR-10.

Research Objectives

01

Reproduce

Implement Jia et al. faithfully from the published description and document every assumption it forces.

02

Improve

Design a defensible architecture: multi-scale local features, learned chunk selection, gated fusion.

03

Encrypt

Implement genuine CKKS inference with a stated threat model and measured overhead.

04

Measure

Accuracy, macro P/R/F1, confusion, ablation, encryption cost — on one identical held-out split.

05

Explain

Grad-CAM and attention maps, including the failure cases.

06

Report honestly

State the dataset ceiling, the negative results, and the limits of the privacy claim.

Dataset · NEU-CLS

Verified against the published specification, not assumed

1800
images, 200×200 grayscale

6
defect classes, 300 each

1
exact duplicate pair found and removed

0
near-duplicate leakage after audit

split	images	per class
train	1223	204, 204, 203, 204, 204, 204
val	216	36, 36, 36, 36, 36, 36
test	360	60, 60, 60, 60, 60, 60

Integrity checks

SHA-256 pixel hashing found one duplicate pair; it is removed before splitting so no image can span train and test.

Dataset Integrity · near-duplicate audit

NEU-CLS patches come from a limited set of source photographs, so pixel-hashing alone is not enough

representation	max sim	≥0.90	same-class	cross-class max
32px	0.990	66	50%	0.984
64px	0.987	58	48%	0.980
128px	0.986	48	44%	0.977
128px high-pass	**0.848**	**0**	**50%**	**0.545**

This matters: a 100% test accuracy would otherwise be indistinguishable from memorising duplicated crops. The audit is what licenses the result.

Conclusion: no leakage

Raw-pixel similarity is high at low resolution, but the best cross-class match is just as high — the signature of shared illumination, not repeated content. With illumination removed, no test image matches any training image above 0.85.

Literature Review

All citations machine-verified against Crossref / arXiv

work	dataset	method	relevance
Song & Yan (2013)	NEU-CLS	Completed LBP	originating dataset paper
Jia et al. (2023)	NEU-CLS	chunk CNN + CKKS	97.1% — base paper
Cheon et al. (2017)	—	CKKS scheme	the encryption we use
Tan & Le (2019)	ImageNet	EfficientNet	our global branch
Hu et al. (2018)	ImageNet	Squeeze-and-Excitation	channel-gating precedent
Lee et al. (2022)	—	minimax polynomial	higher-degree alternative to square
Elkhatib et al. (2025)	CIFAR-10	PRISM, CKKS + poly	closest prior art on boundary design

15 of 15 citations resolved to verifiable publisher metadata (scripts/verify_citations.py → reports/citations_verified.json). Works that could not be verified are listed in the report but are not cited as evidence.

Existing Approaches · the benchmark is saturated

This single fact shapes the entire experimental design

— Published NEU-CLS results cluster at 99%+: shallow attention CNNs, wavelet texture descriptors and modern backbones all report near-ceiling accuracy.

— We reproduce that ceiling directly: a pretrained EfficientNet-B0 reaches 100% test accuracy on our split within two epochs.

— At ceiling, every architectural variant scores identically. An ablation run at full training size would rank noise and present it as evidence.

Consequence for this project

Component contributions are measured in a reduced-data regime (15% of the training split, ~30 images per class), where the objective still has dynamic range. Full-data numbers are reported too — and reported as a ceiling, not as a ranking.

This is the difference between an ablation that measures something and an ablation that decorates a conclusion.

Base Paper · Jia et al. (2023)

Journal of Cloud Computing 12:175 · DOI 10.1186/s13677-023-00537-0 · CC BY 4.0

— **Preprocess.** resize 112×112, grayscale, global histogram equalisation.

— **Chunk.** 28×28 subregions → a 4×4 grid of 16 chunks.

— **Select.** Sobel gradient per chunk; keep those above a threshold.

— **Encrypt.** CKKS on the retained chunks, polynomial modulus 16384.

— **Classify.** 7×7 conv → FC 256→128 → FC 128→6; square activation under encryption.

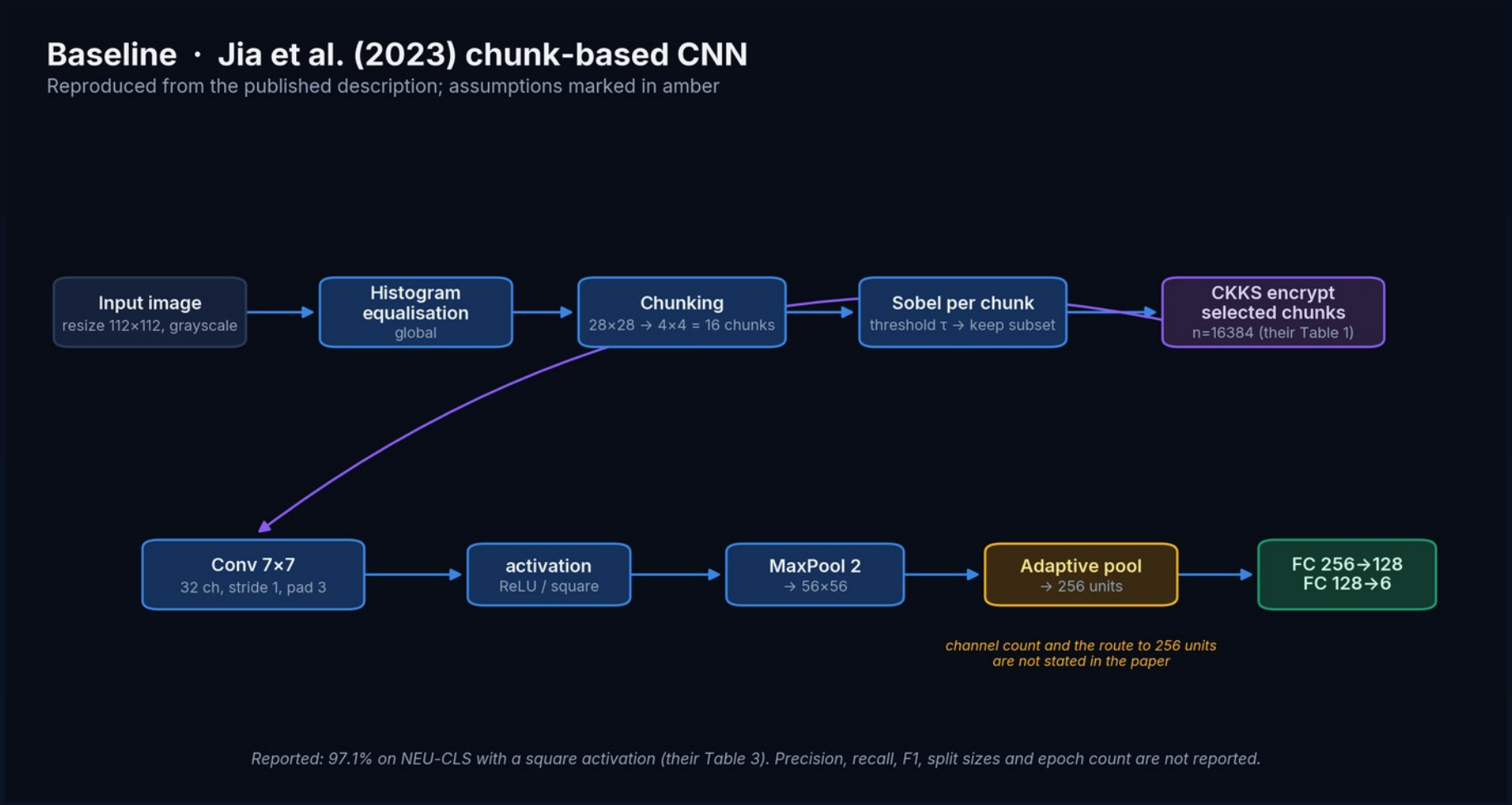
97.1%
reported NEU-CLS accuracy (Table 3)

235 ms
encrypt feature chunks only (Table 2)

806 MB
Galois keys at n=16384 (Table 1)

Base Paper Architecture

Reproduced; amber marks what the paper does not state



Base Paper · limitations found on close reading

Stated to scope a fair comparison, not to disparage the work

Cross-dataset comparison

Table 3 ranks their NEU-CLS result against TCGA genomics, Fashion-MNIST and CIFAR-10.

No P / R / F1 for NEU-CLS

Accuracy only; NEU-CLS results appear as figures, not tables.

Protocol unreported

Split sizes, epochs, learning rate and batch size are absent.

Chunk size inconsistent

28×28 in one section, 28×26 in two others; only 28×28 fits Table 2.

CKKS expanded incorrectly

“Complex Key-Value Storage” in one section.

Encryption improves accuracy

Reported, but CKKS is approximate — it adds error, not signal.

01

Research Gap and Proposed Solution

From a fixed photometric threshold to a learned, measured selection

Research Gap

dimension	Jia et al. (2023)	SAGE-HE
chunk selection	fixed threshold $\|\nabla\| > \tau$	learned score + Sobel prior, λ measured
spatial scales	single 7×7 receptive field	1×1 / 3×3 / dilated / 5×5 in parallel
branch combination	none (single path)	channel-wise gated fusion
what is encrypted	selected image chunks	compressed 64-d feature vector
evaluation	accuracy only	macro P/R/F1, confusion, CI, ablation
comparison	across different datasets	same held-out split for both models

Every row is a testable claim

Each line maps to a component slide and to a numbered ablation stage, so none of them has to be taken on trust.

Spoiler, stated up front

The last two rows survive contact with the evidence. Most of the first four do not — the ablation later in this deck is where that is shown.

Proposed Solution · SAGE-HE

Secure Adaptive Gated Ensemble with Homomorphic Encryption

— **Global branch.** EfficientNet-B0, ImageNet-pretrained, adapted to one input channel.

— **Local branch.** Multi-scale CNN with four parallel receptive fields per block.

— **Adaptive selection.** Chunk attention scored by a learned MLP plus a Sobel prior weighted by a learnable λ.

— **Gated fusion.** A per-channel gate z mixes the branches: $z \cdot g + (1-z) \cdot l$.

— **Compression.** 256 → 64, tanh-bounded — the vector that gets encrypted.

— **Encrypted head.** $W_2 \cdot \text{square}(W_1 v + b_1) + b_2$ evaluated under CKKS.

Design constraint that drives everything

Only a depth-3 polynomial is affordable under CKKS. So the network is split: arbitrary operations client-side, polynomial-only server-side.

Trained as deployed

The head is trained with the square activation it will be evaluated with — not trained with ReLU and swapped afterwards.

The next slides explain each component and why it was expected to help. Treat them as hypotheses: the ablation later in this deck tests every one of them against the bare backbone, and most do not survive.

Novelty and Contributions

Stated narrowly, because the surrounding literature is dense

NOT CLAIMED AS NOVEL

— Splitting a network so only a polynomial head is encrypted — this is PRISM (2025) and others.

— The square activation for CKKS — the base paper and CryptoNets lineage.

— Dual-branch global/local designs, or gating as a mechanism.

— EfficientNet on NEU-CLS.

WHAT ACTUALLY HOLDS UP

— A reproduction and audit of the base paper it does not provide: macro metrics, confusion, CIs, and eight documented under-specifications.

— Genuine CKKS inference with a quantified exactness bound, not just an agreement count.

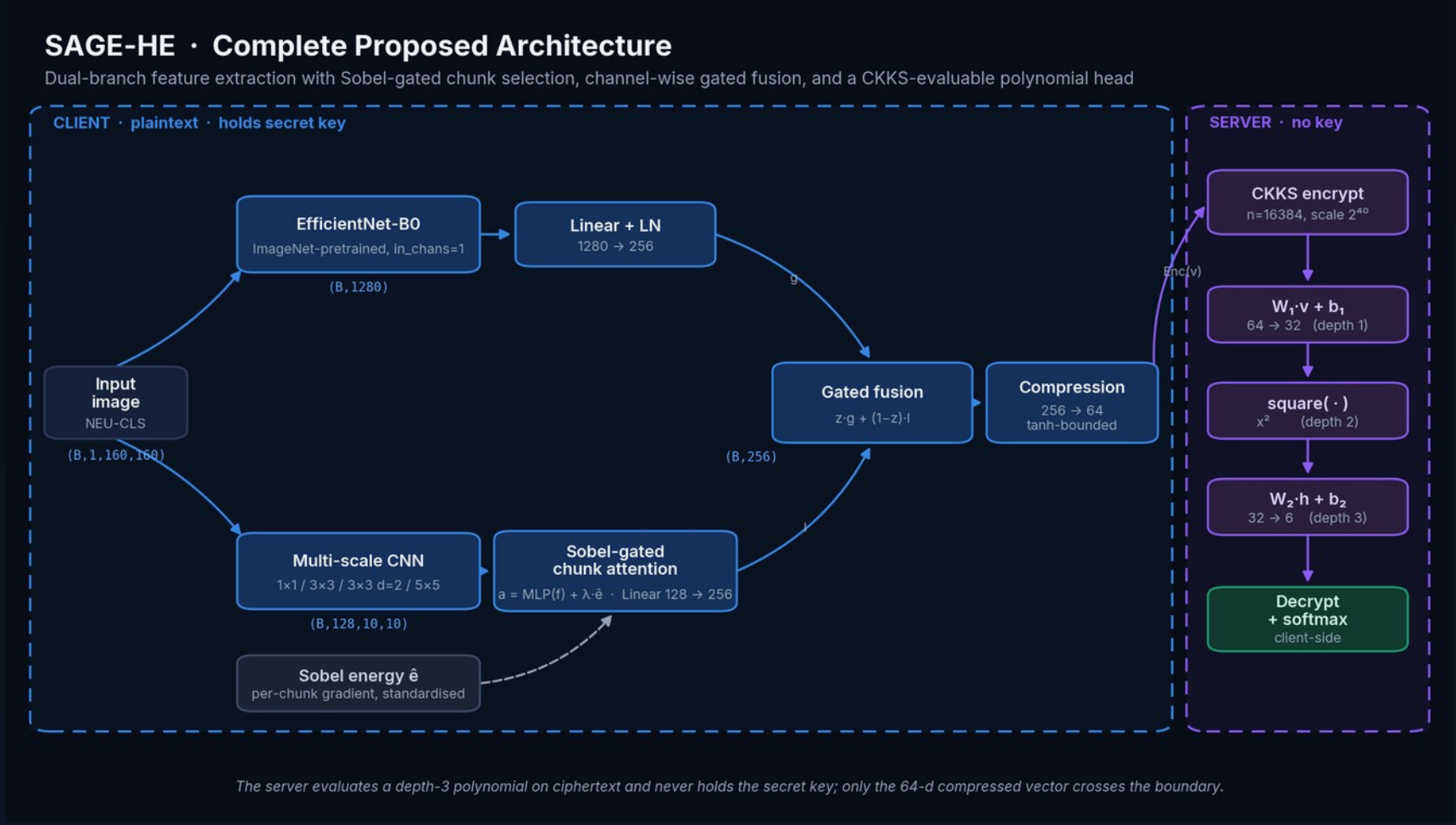
— An evaluation protocol that stays informative on a saturated benchmark — and that is what exposed the result below.

— An explicit threat-model comparison that states what is NOT protected.

The proposed architecture is NOT among the contributions

The learned Sobel-conditioned chunk selection was the one genuinely new piece. Measured against its own backbone under matched seeds, it costs accuracy. We report that rather than claim an architectural advance — see the ablation.

Complete Proposed Architecture



Data Preprocessing

CLAHE replaces the base paper's global equalisation — a choice the design study measures

Dataset Preprocessing Pipeline

Deterministic for validation and test; stochastic augmentation on train only



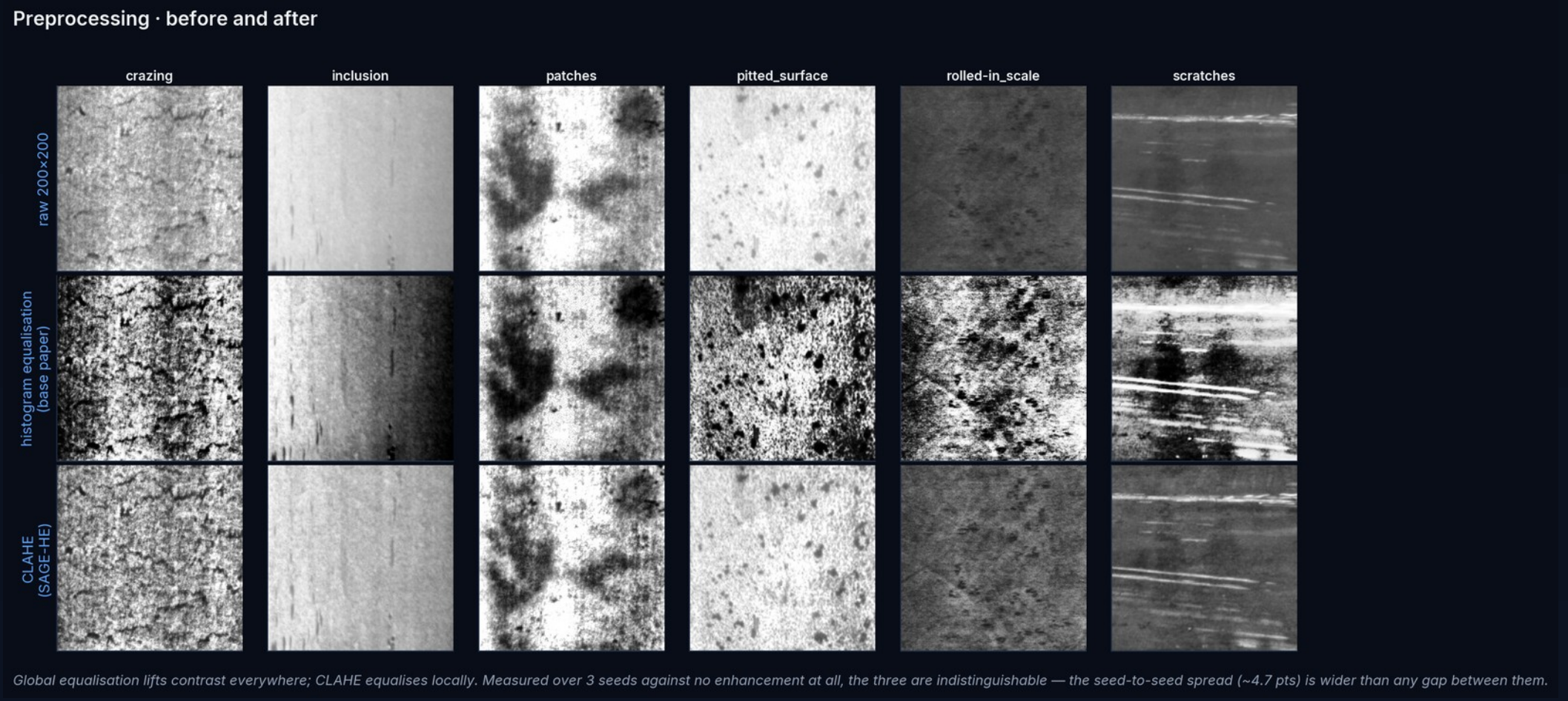
CLAHE vs global equalisation
measured — neither wins

Why no vertical flip
scratches / rolled-in scale carry a horizontal orientation prior

Contrast enhancement is applied to the raw uint8 image before any geometry, so that reflect-padding from rotation cannot bias the equaliser's histogram. The design study measures CLAHE, global equalisation and no enhancement; no enhancement scores highest.

Preprocessing · before and after

The same six images under the base paper's contrast step and ours



Global equalisation lifts contrast everywhere; CLAHE equalises locally. Measured over 3 seeds against no enhancement at all, the three are indistinguishable — the seed-to-seed spread (~4.7 pts) is wider than any gap between them.

Global Feature Extraction · EfficientNet-B0

Pretrained weights, adapted to a single-channel input

— **Why EfficientNet-B0.** Compound scaling gives strong ImageNet features at ~4.0 M parameters — the accuracy/compute point that fits a CPU-trained project.

— **Grayscale adaptation.** timm's in_chans=1 sums the pretrained RGB stem filters rather than replicating the channel, preserving filter response magnitude.

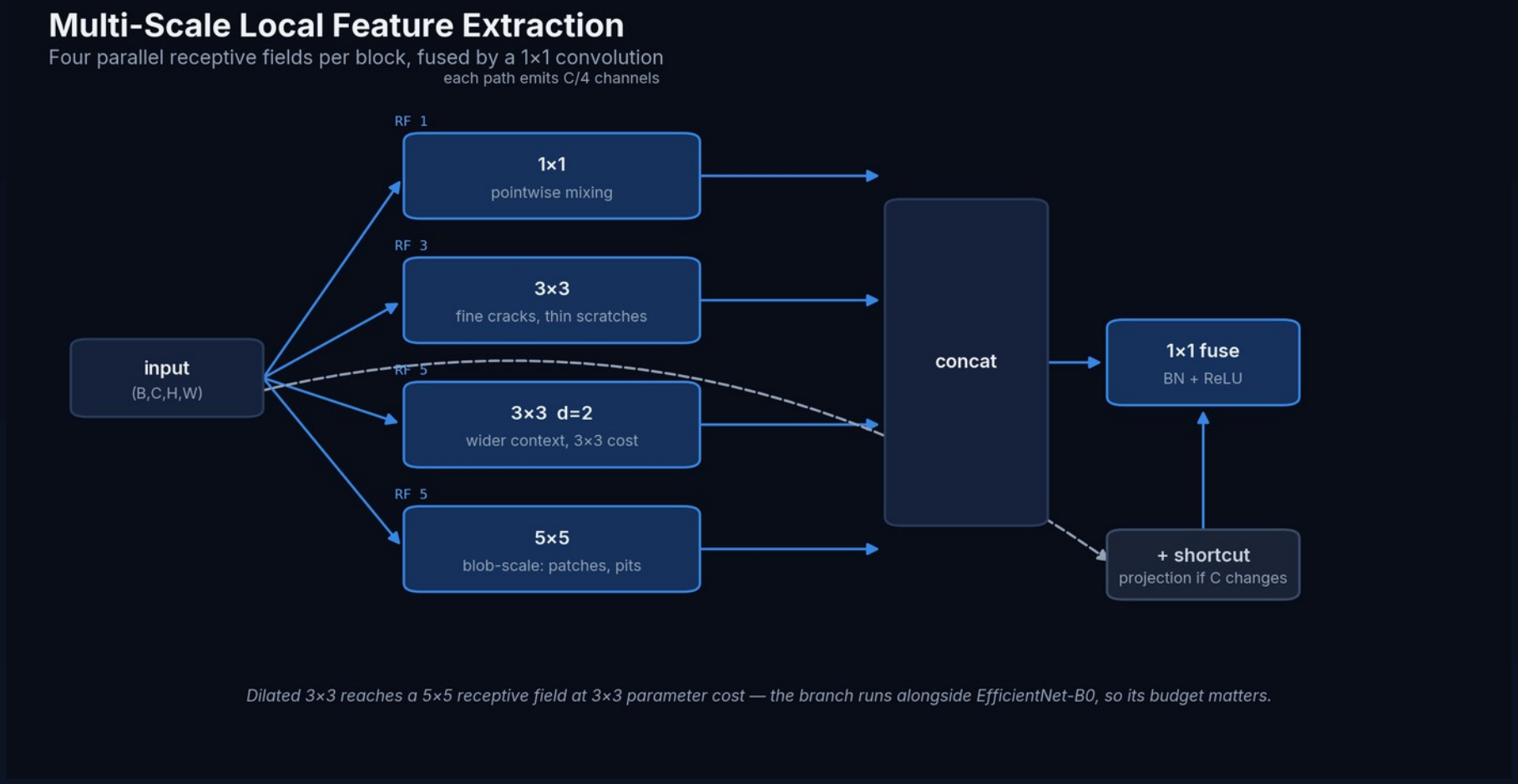
— **Projection.** 1280-d pooled features → Linear + LayerNorm → 256-d, matching the local branch.

— **Transfer matters most when data is scarce.** This is visible in the learning curve: the gap over a from-scratch CNN widens as training data shrinks.

component	parameters
global branch	4,335,420
local branch	190,816
chunk attention	8,322
fusion	197,120
compression	16,448
encrypted head	2,278
TOTAL	**4,750,404**

Multi-Scale Local Branch

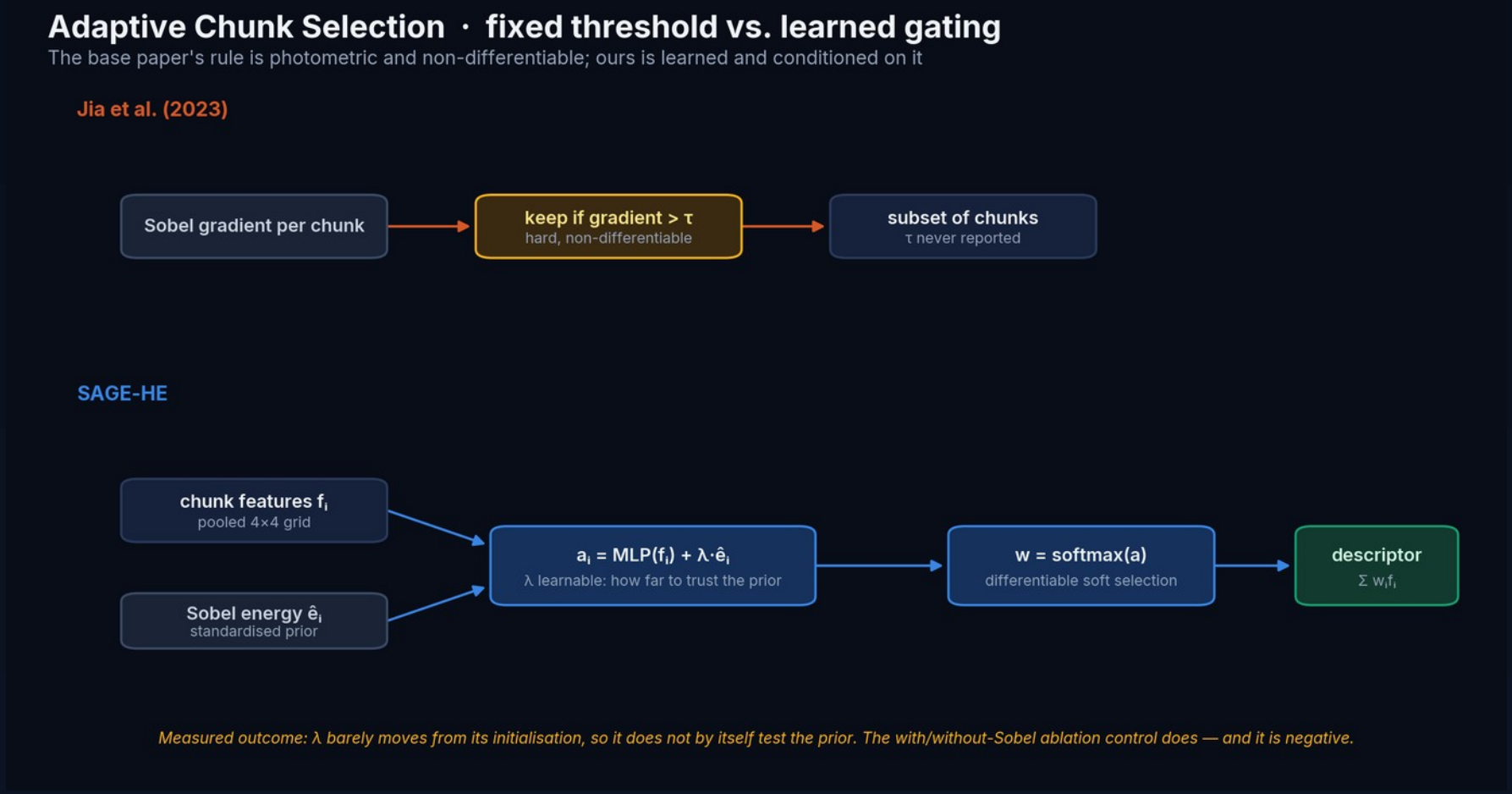
Four parallel receptive fields per block, fused by a 1×1 convolution.



Dilated 3×3 reaches a 5×5 receptive field at 3×3 parameter cost — the branch runs alongside EfficientNet-B0, so its budget matters.

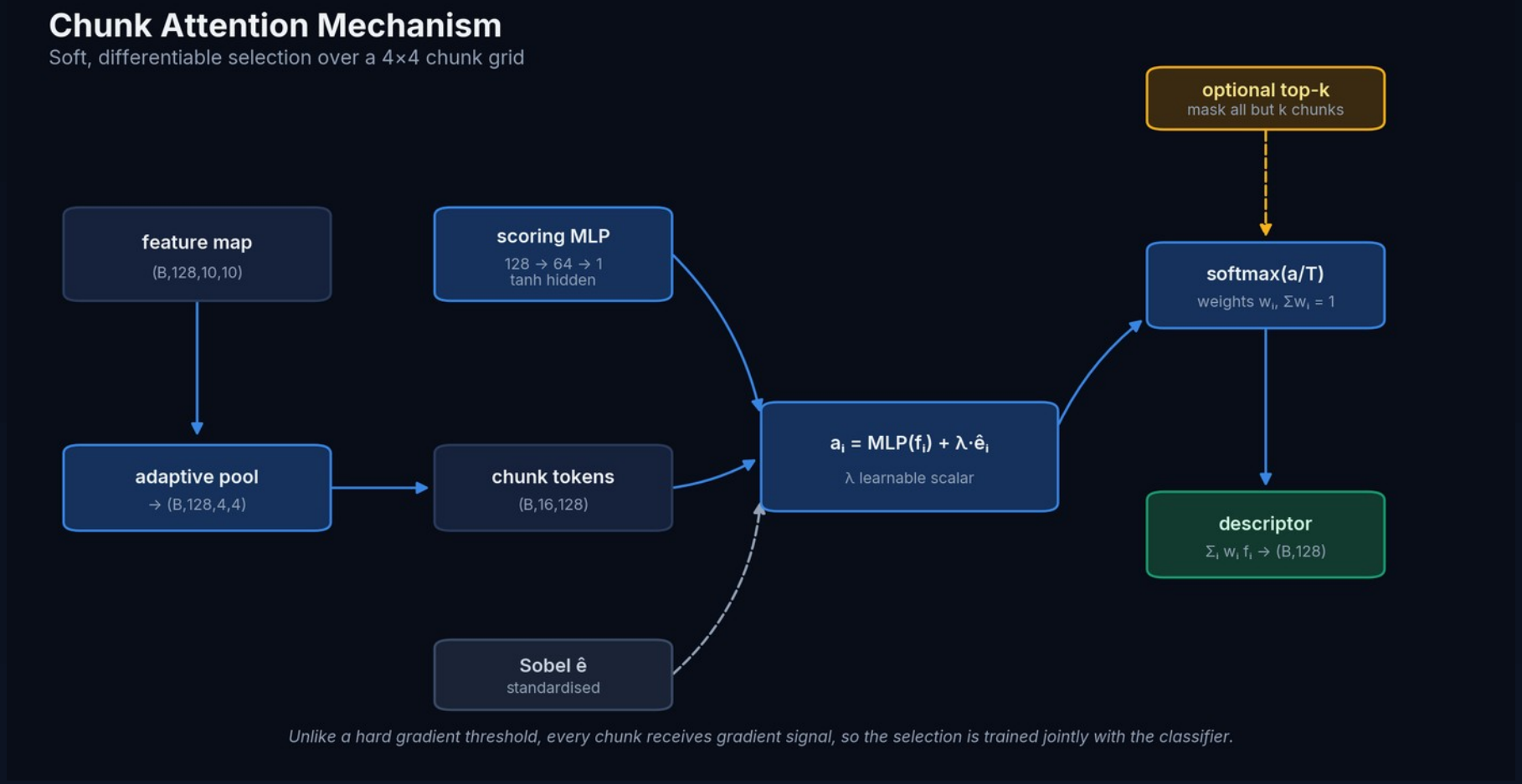
Adaptive Chunk Selection

The base paper's rule is photometric and non-differentiable; ours is learned and conditioned on it.



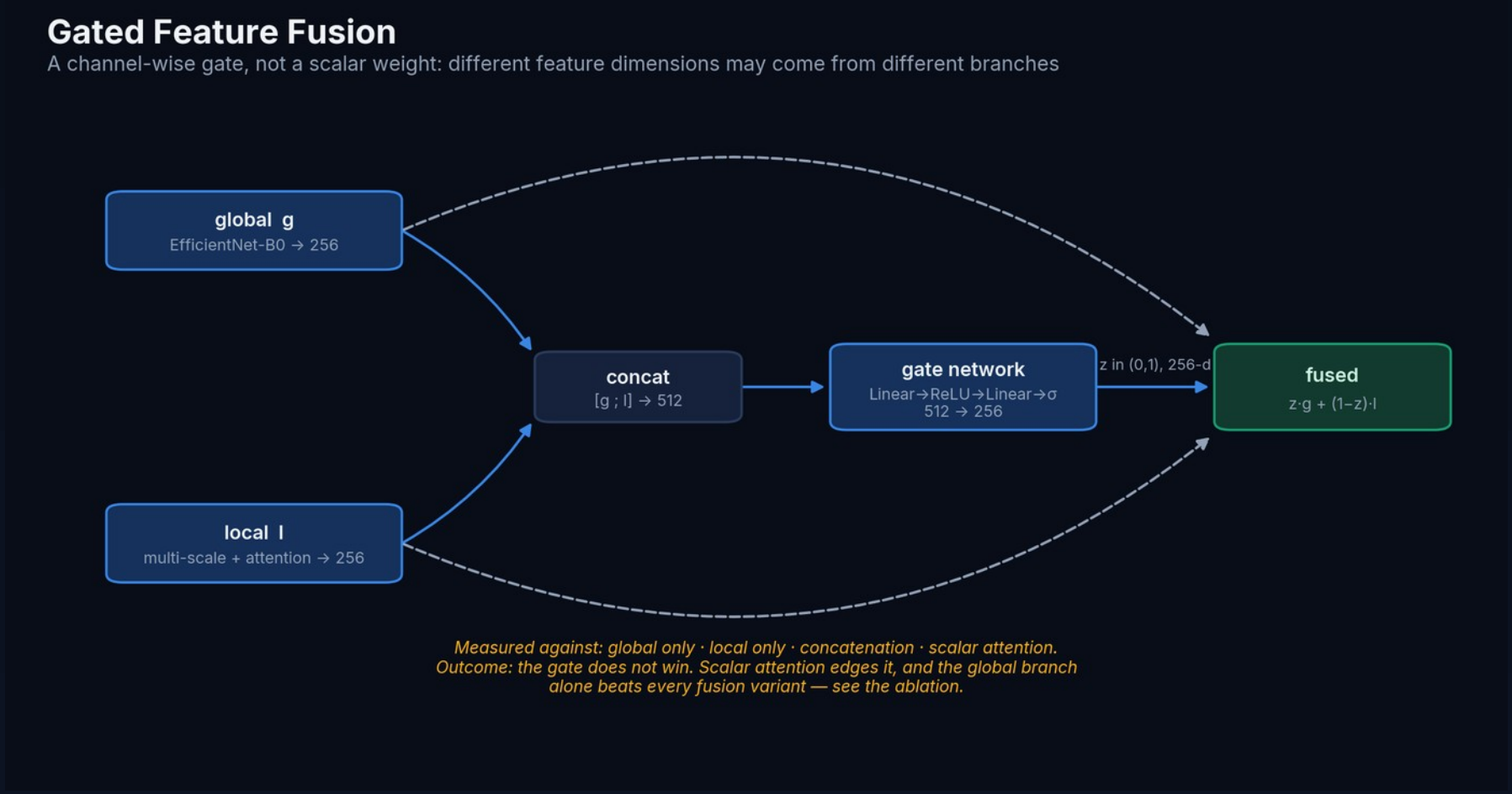
Chunk Attention Mechanism

Soft, differentiable selection over the 4×4 chunk grid.



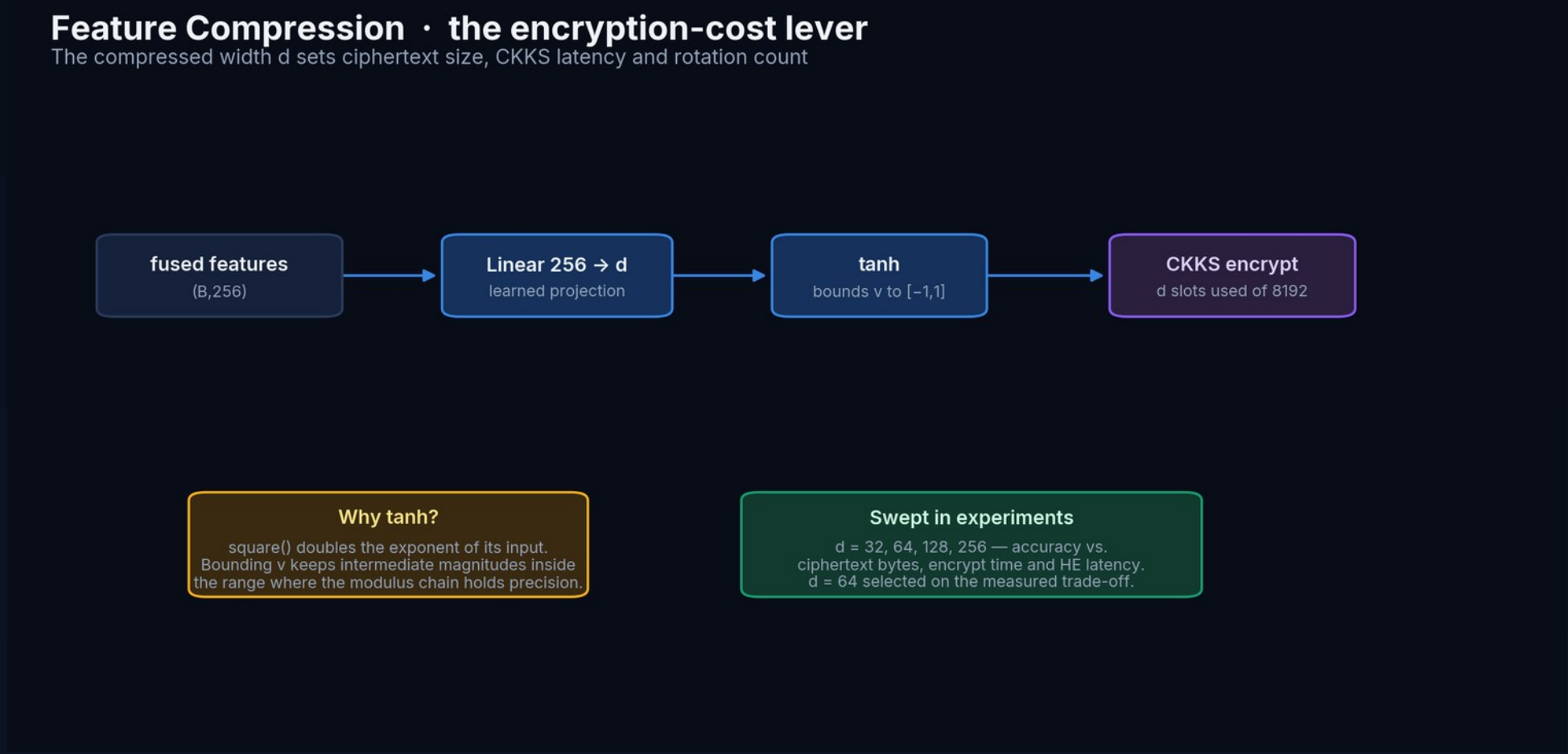
Gated Feature Fusion

A channel-wise gate, not a scalar weight.



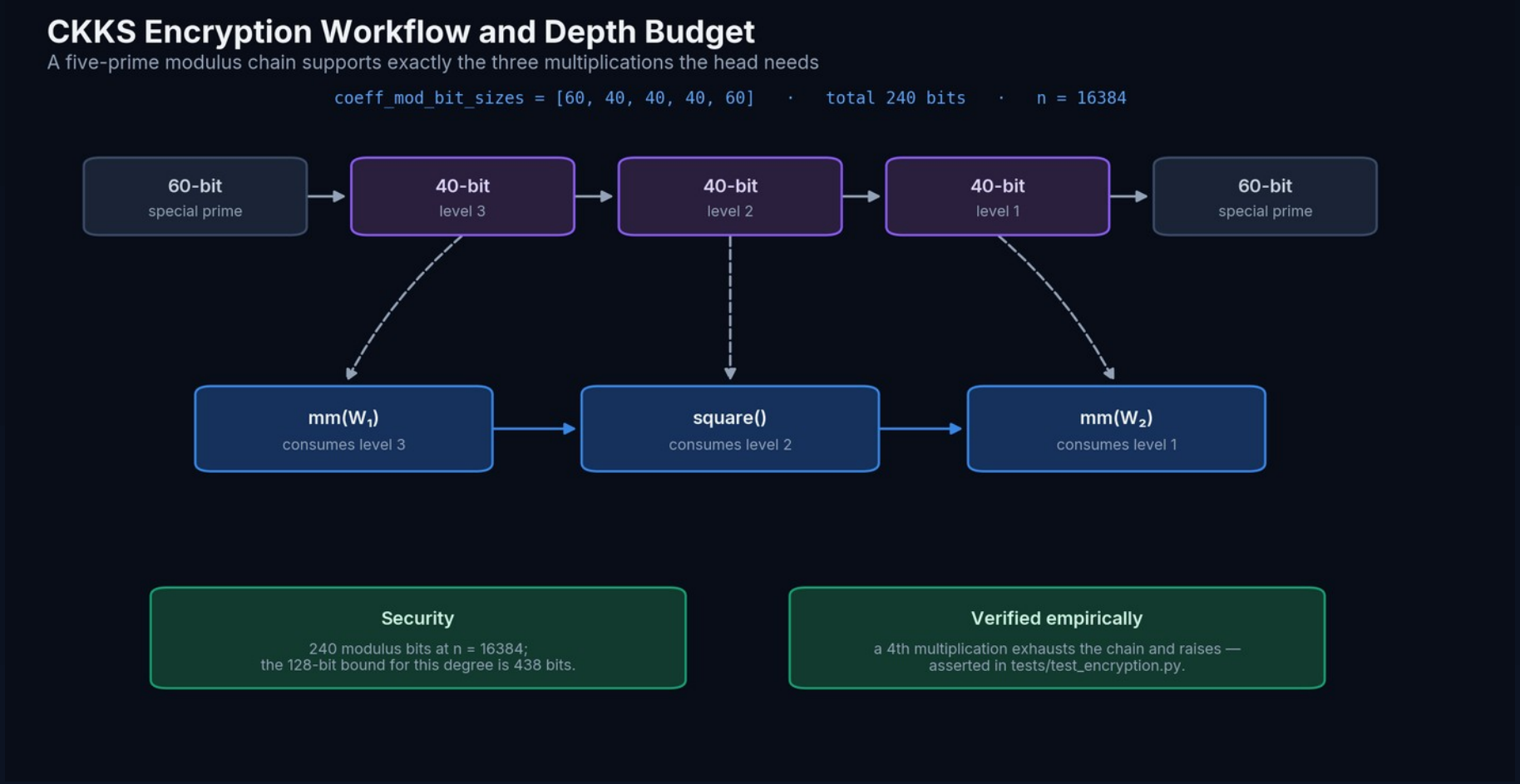
Feature Compression

The compressed width d is the encryption-cost lever.



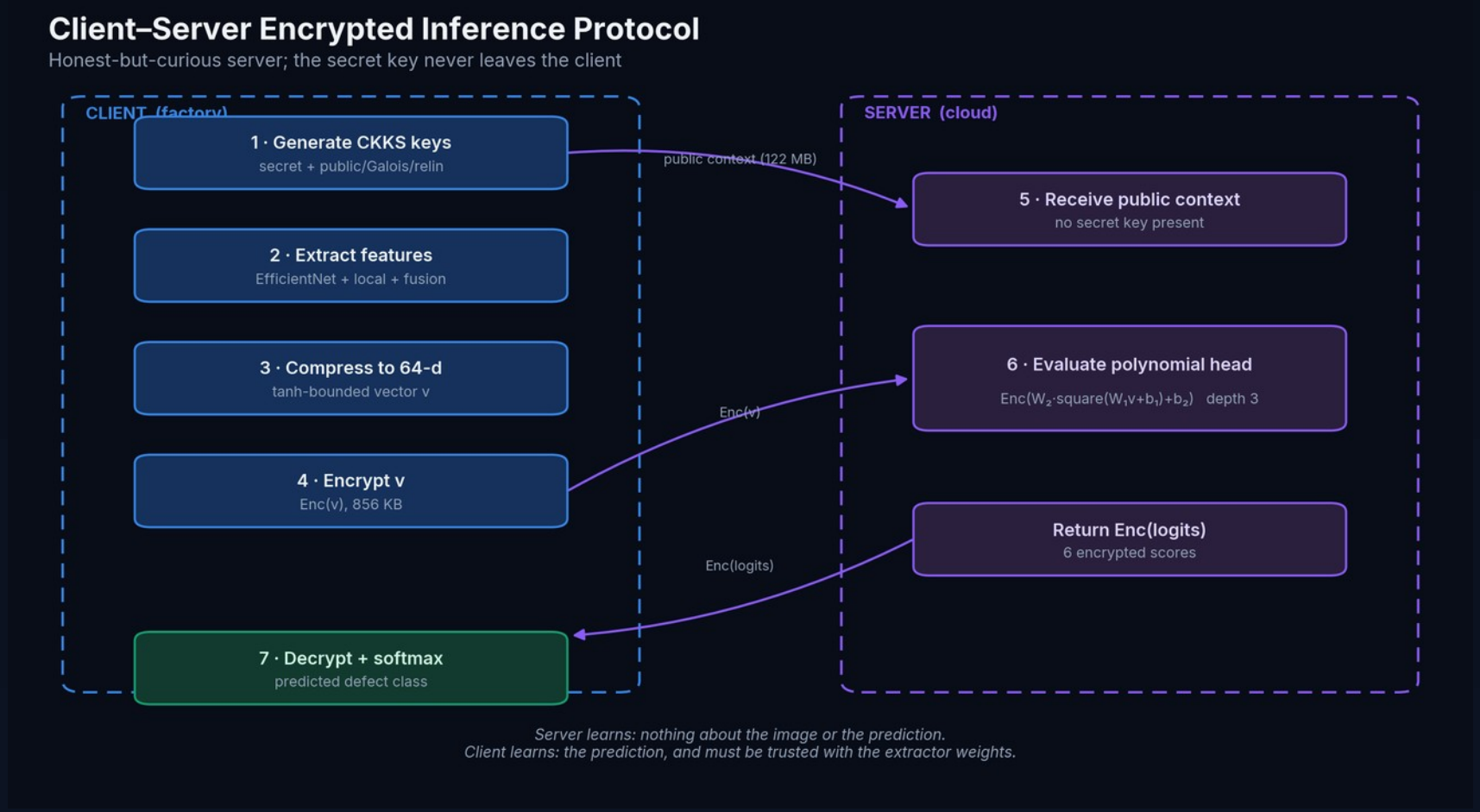
CKKS Encryption

A five-prime modulus chain supports exactly the three multiplications the head needs.



Encrypted Inference Workflow

Honest-but-curious server; the secret key never leaves the client.



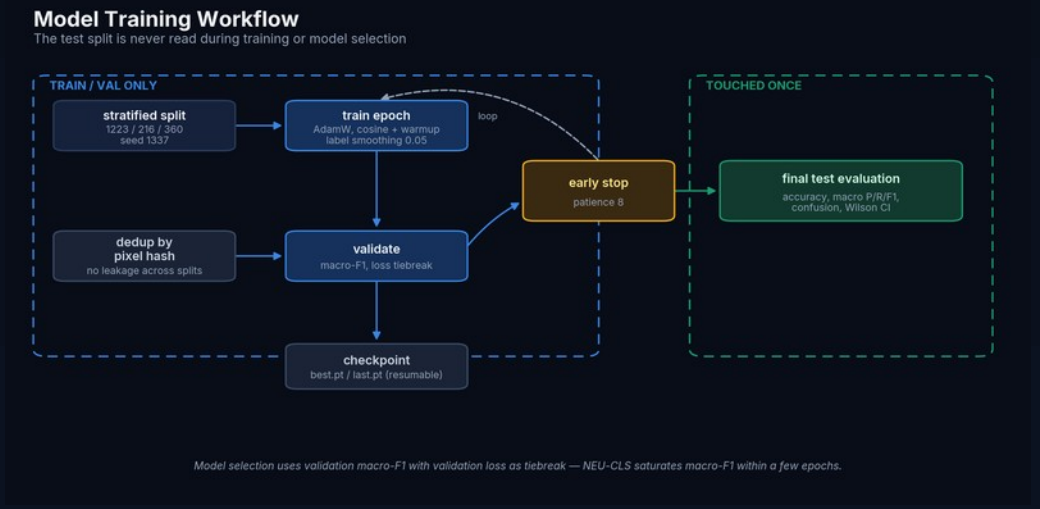
Training Methodology

— **Optimiser.** AdamW, cosine schedule with linear warm-up, gradient clipping at 5.0.

— **Loss.** Cross-entropy with 0.05 label smoothing.

— **Selection.** Validation macro-F1, with validation loss as tiebreak — NEU-CLS saturates macro-F1 within a couple of epochs, so the tiebreak is what picks a well-calibrated model rather than the first lucky epoch.

— **Hygiene.** Fixed seeds, saved split manifests, resumable checkpoints; the test split is read exactly once, after selection.



Experimental Configuration

Identical data and protocol for both models

setting	value
dataset	NEU-CLS, 1799 images after deduplication
split	1223 train / 216 val / 360 test, stratified, seed 1337
seeds	3 independent training seeds per architecture
SAGE-HE input	160×160, CLAHE
baseline input	112×112, histogram equalisation (as published)
hardware	4-vCPU x86-64, no GPU — all timings are CPU timings
CKKS	n=16384, [60,40,40,40,60], scale 2^{40}, depth 3

No GPU was used

The project brief specifies GPU training. None was available, so every model was trained on CPU and every latency is a CPU latency. Reported rather than papered over; the code selects CUDA automatically where present.

Hyperparameter Optimisation · Optuna

Run in the reduced-data regime, where the objective still has dynamic range

15	0.9862	10%	8
TPE trials	best validation macro-F1	training data per trial	epochs per trial

parameter	best value
contrast	histeq
lr	0.0005304883441224623
weight_decay	1.2242594760084797e-06
batch_size	16
compressed_dim	64
head_hidden	64
fusion	gated
use_sobel	False

parameter	importance
lr	0.899
use_sobel	0.039
contrast	0.025
fusion	0.017
head_hidden	0.013
compressed_dim	0.006

The search could pick the fusion strategy, the compressed width, the head width, the contrast step and whether to use the Sobel prior at all. It puts 0.899 of the importance on `lr` alone and 0.099 on every architectural choice combined — and its best trial switches the Sobel prior off. A completely different procedure, pointing the same way as the ablation.

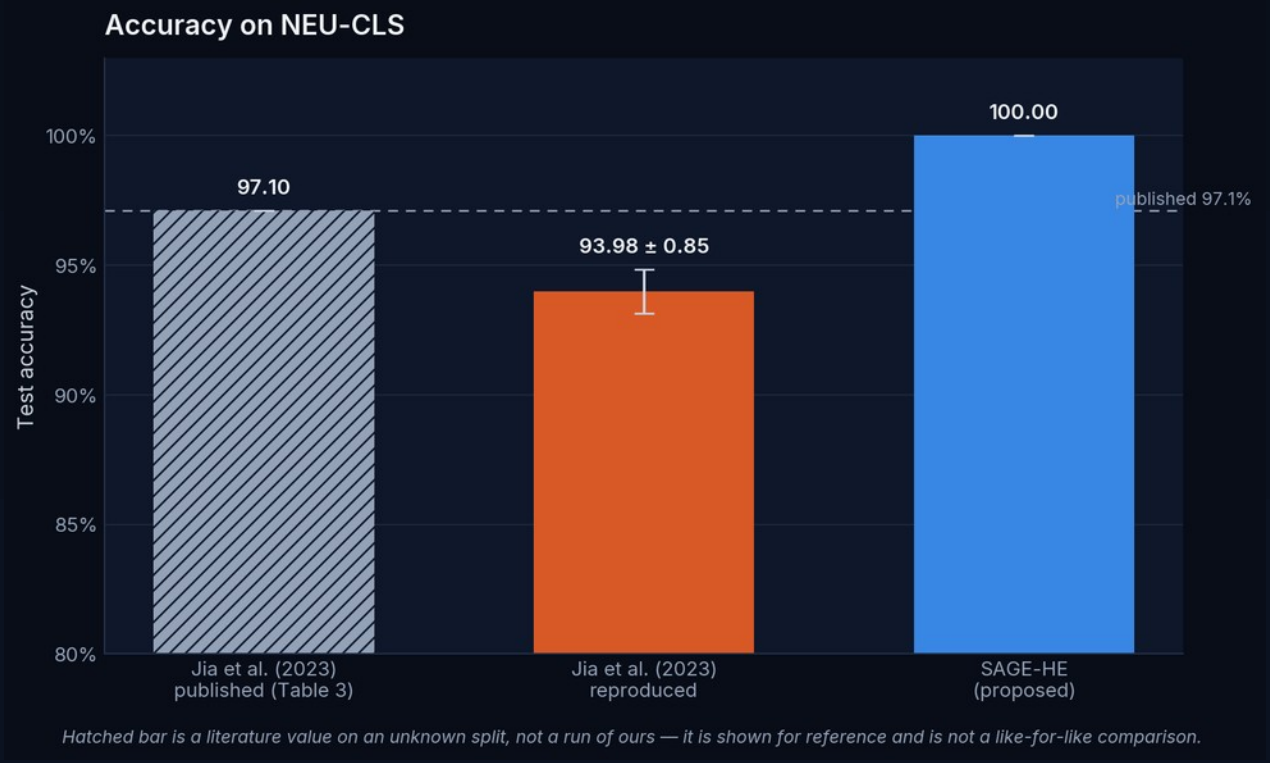
02

Experimental Results

One held-out test split, read once, for every number that follows

Accuracy Comparison

Mean ± standard deviation over 3 seeds, identical held-out split



Read this carefully

SAGE-HE beats our reproduced baseline by 6.0 points. It does NOT follow that it beats the published 97.1%, which was measured on an unknown split.

Reproduction finding

Our faithful reproduction reaches 93.98%, below the published 97.1%. We could not reproduce the paper's figure from its own description — reported as a result in its own right.

This margin is NOT evidence for the proposed components. SAGE-HE uses a pretrained 4.7 M-parameter backbone; the baseline is a 35 k-parameter network trained from scratch, because that is what the paper describes. Transfer learning and capacity explain much of the gap — the architectural contribution is isolated in the ablation, where every stage shares the same backbone.

Did the reproduction fail because of OUR assumptions?

The 97.1% claim rests on choices the paper never states — so we attacked our own finding

baseline variant (seed 42, full data)	accuracy %
16 channels, square	95.00
32 channels, square (our reproduction)	94.72
64 channels, square	90.28
32 channels, ReLU	86.94
128 channels, square	86.67
published claim (Jia et al., Table 3)	**97.10**

The finding survives the attack

Varying the two things the paper leaves unspecified — convolution width and activation — the best variant reaches 95.00%, still 2.10 points short of the published figure.

So the gap is not an artefact of our assumptions. Some other unreported detail may explain it; we cannot test what is not described.

A side finding

Swapping to ReLU costs 7.8 points against the square activation. On this architecture the CKKS-compatible choice is also the better-performing one — the opposite of the usual 'polynomial activation is the price of encryption' trade-off.

Is the difference statistically real?

Both models scored on the identical 360 test images, so the comparison is paired

+5.28 pts
accuracy improvement

+3.1 to +7.8
95% bootstrap CI on the delta (pts)

p = 3.8e-06
McNemar, exact binomial

19 : 0
images gained : lost

outcome on a test image	count
both correct	341
SAGE-HE only correct	19
baseline only correct	0
both wrong	0
discordant pairs	19

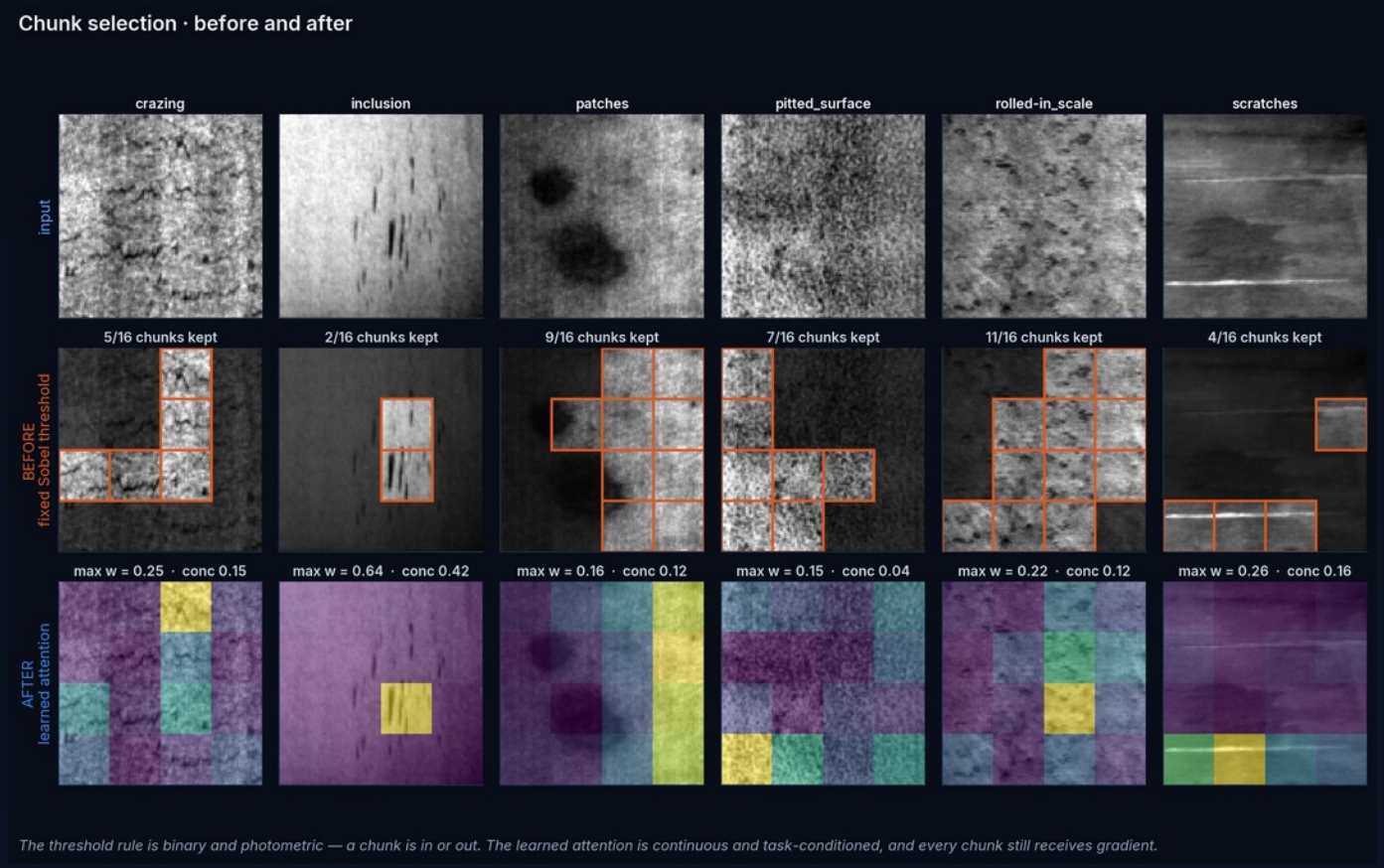
Why McNemar and not a two-sample test

The two models see the same images, so the predictions are paired. McNemar looks only at the images where they disagree and asks whether the split is more lopsided than a coin flip.

Here every one of the 19 disagreements went the same way: SAGE-HE never missed an image the baseline got right.

Chunk Selection · before and after, on real test images

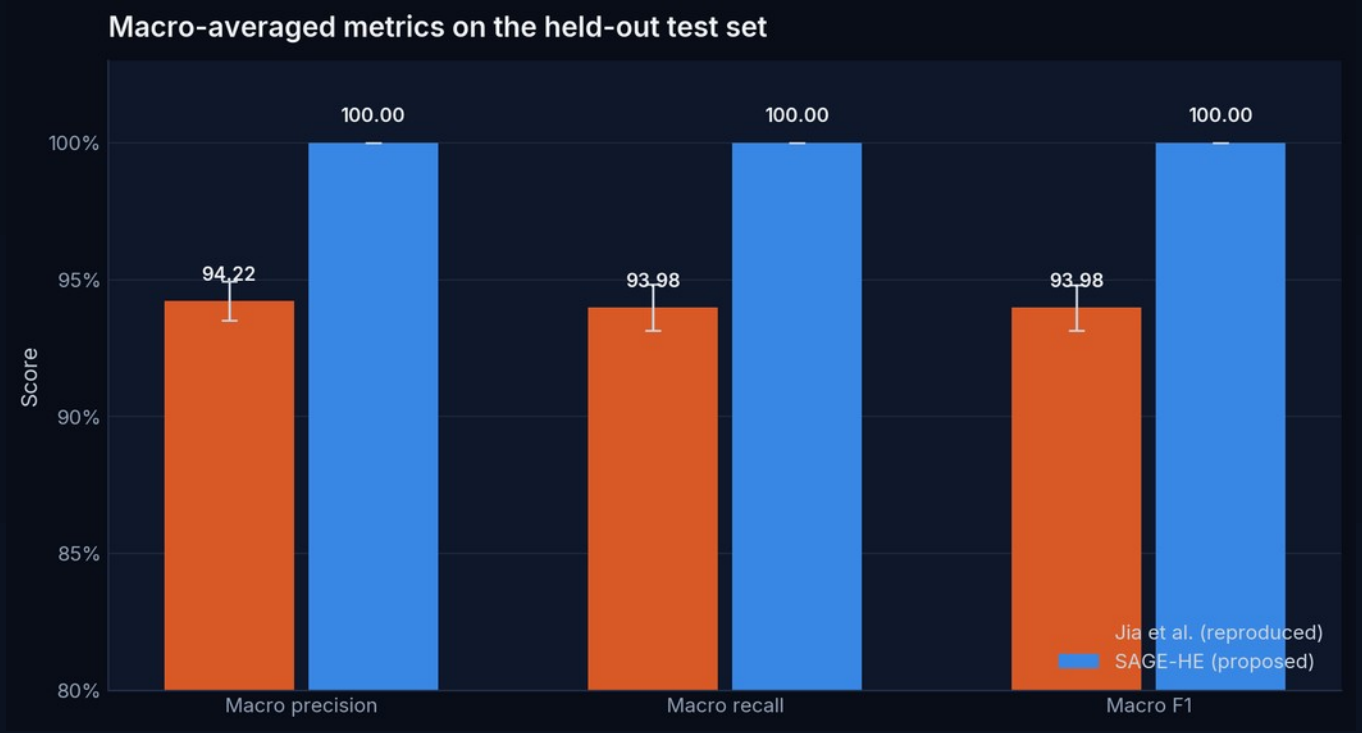
Fixed Sobel threshold (base paper) against the learned attention, same images



The attention is not uniform noise: it concentrates on localised defects (inclusion, concentration 0.42) and spreads out on diffuse texture (pitted_surface, 0.04) — the behaviour a useful selector should show.

Precision and Recall

Macro-averaged — the base paper reports none of these for NEU-CLS



100.00%
SAGE-HE macro precision

100.00%
SAGE-HE macro recall

100.00%
SAGE-HE macro F1

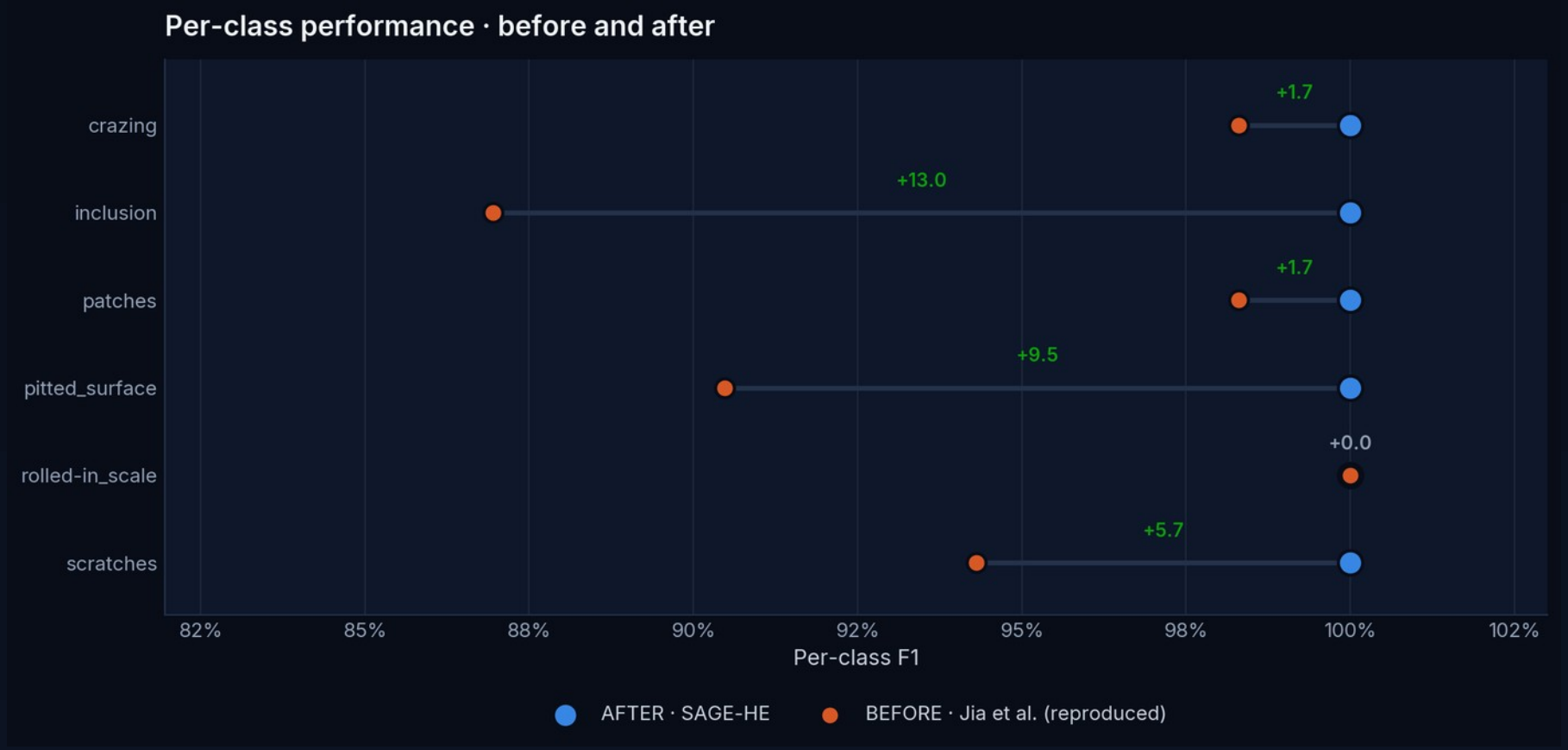
Confusion Matrices · before and after

Identical 360 test images, identical colour scale

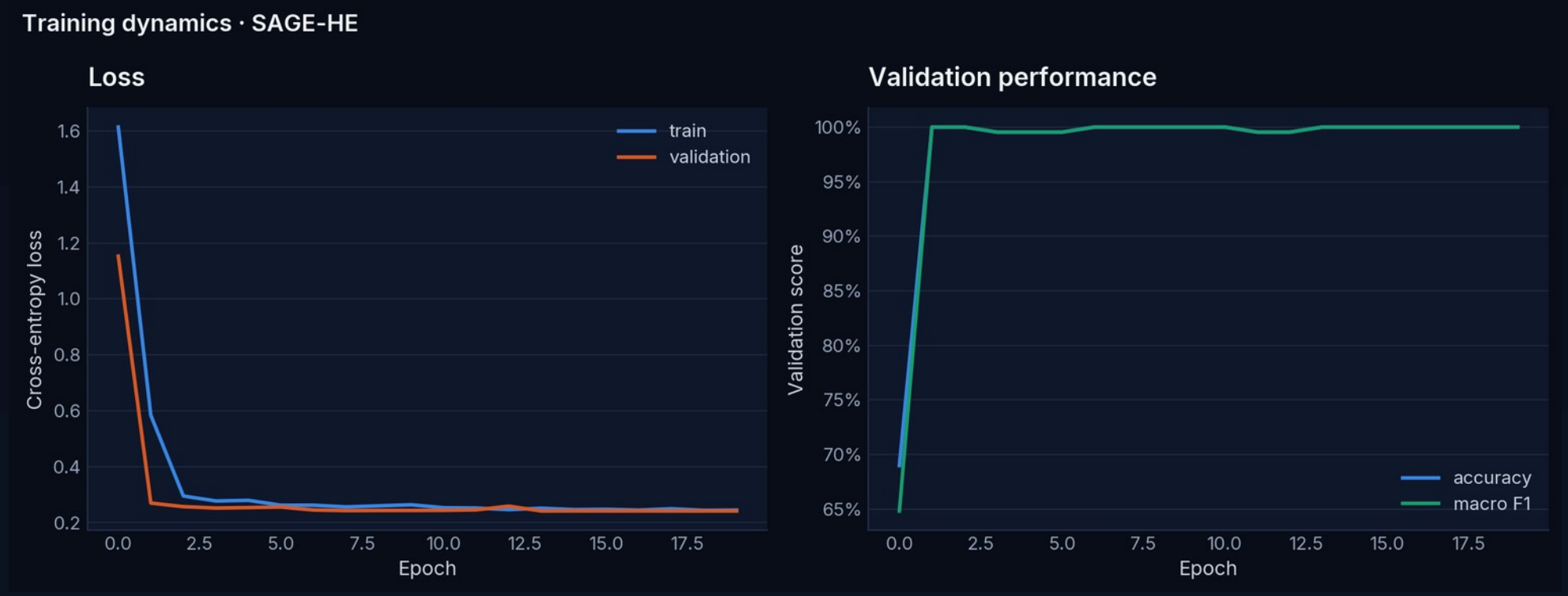


Per-Class Performance · before and after

Per-class F1 for every defect type, same test split



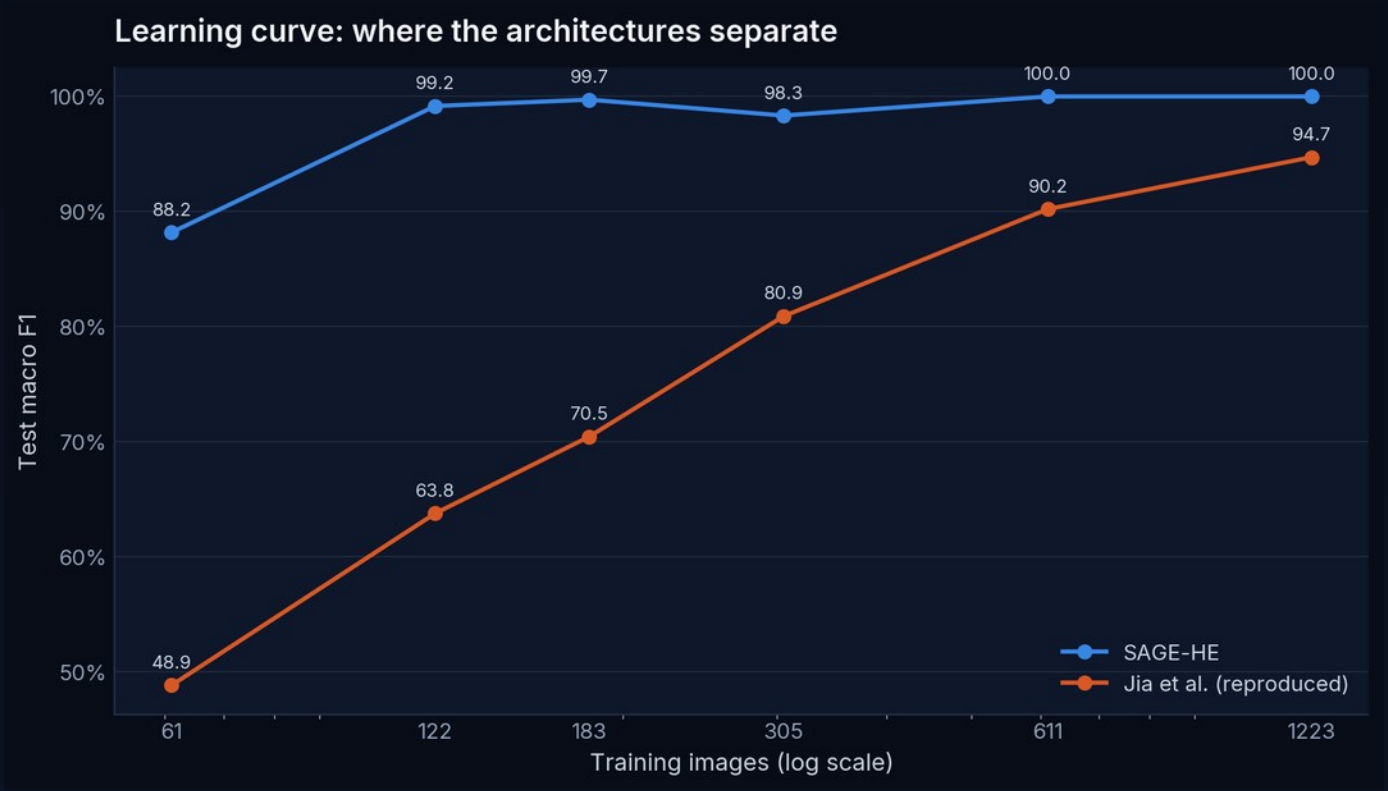
Training and Validation Curves



Validation macro-F1 reaches 1.0 within two epochs — the saturation that motivates the reduced-data ablation protocol. Validation loss keeps improving afterwards, which is what the selection tiebreak uses.

Learning Curve · where the architectures actually separate

Test macro-F1 against training-set size, same held-out test split throughout

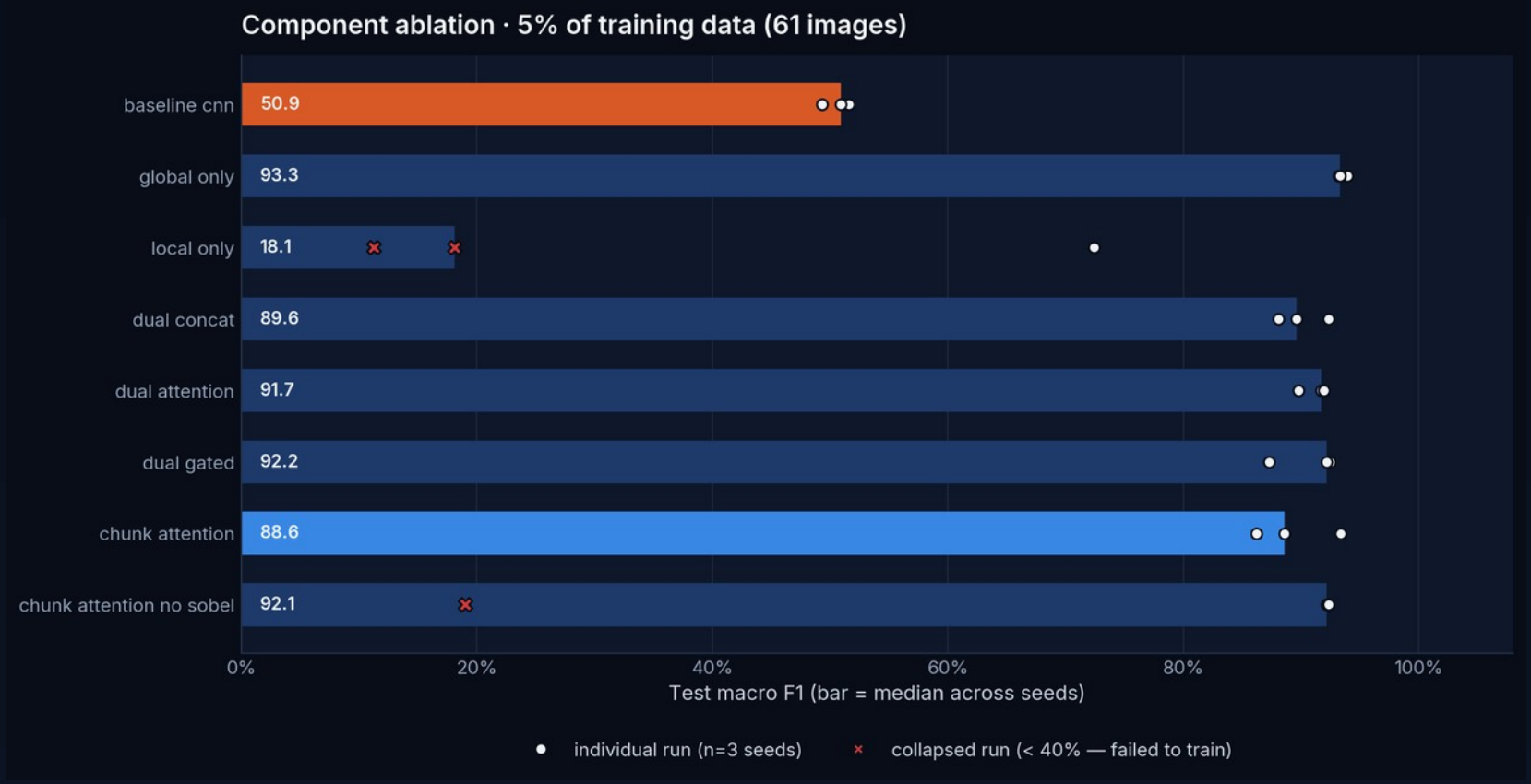


Why this is the honest comparison

At full data both models are near ceiling and the difference is within noise. The gap is real and large in the low-data regime — which is also the regime where a factory has few labelled defect examples.

Ablation Study

Reduced-data regime (15% of training split) so components are separable



Ablation · what the numbers say

configuration	macro F1 (mean ± sd)	accuracy
baseline cnn	50.6 ± 1.2	51.6
global only	93.5 ± 0.4	93.5
local only	33.9 ± 33.5	40.0
dual concat	90.0 ± 2.1	90.0
dual attention	91.1 ± 1.2	91.2
dual gated	90.6 ± 2.9	90.9
chunk attention	89.4 ± 3.6	89.6
chunk attention no sobel	67.8 ± 42.3	70.6

Every proposed component costs accuracy

adding the local branch -2.82 pts

adding chunk attention -1.26 pts

adding the Sobel prior -1.28 pts

full model vs its backbone -4.08 pts

What is actually established

Two contrasts hold on every seed and carry the conclusion: the local branch costs accuracy, and the full model scores below its own backbone (−0.6 to −7.0 pts, negative on all three seeds).

Finer attributions — chunk attention vs. Sobel prior — do not separate cleanly at this sample size and are not claimed.

Design Studies

Each variant changes one choice against the specified configuration (89.38% macro F1)

INPUT RESOLUTION

variant	macro F1 %
112	87.4 ± 5.4
200	91.8 ± 2.0

CONTRAST ENHANCEMENT

variant	macro F1 %
histeq	87.9 ± 4.3
none	91.2 ± 4.7

COMPRESSED DIMENSION

variant	macro F1 %
128	91.1 ± 1.8
256	90.7 ± 1.9
32	56.9 ± 39.6

HARD TOP-K SELECTION

variant	macro F1 %
4	43.0 ± 41.9
8	90.5 ± 2.9

Only 2 of 9 variants differ from the reference by more than their own seed-to-seed spread — the rest are indistinguishable, not ranked. Measured at 61 training images: at full training size every variant sits at ceiling and these studies would show nothing, which is the same reason the ablation is run here.

What did not work

Reported because the evidence says so, not because it helps the story

λ never learned anything

The learnable Sobel trust was meant to be a readout: drive it to zero and the model is telling us the prior is useless.

It moves by at most 0.0097 from its initialisation of 1.0 in every run. One unregularised scalar among 4.7 M parameters gets negligible gradient.

Training collapses at 61 images

Some low-data configurations fail to train entirely — the no-Sobel control scored 19% macro F1 on one seed and 92% on another. Collapsed runs are flagged and excluded from the robust deltas rather than averaged in, because one collapse inverts the sign of a three-seed mean.

BACKBONE ALONE vs THE FULL MODEL, MATCHED SEEDS

training data	backbone	full model	Δ	per-seed
5%	93.46	89.38	−4.08	all negative
10%	98.81	99.35	+0.55	sign flips

The effect is regime-dependent

Clearly harmful when labelled data is scarce; indistinguishable once it is not, as both arms reach the dataset ceiling. No measured regime shows a reliable benefit.

What did not work (continued)

Mechanism, and why the architecture still gets documented in full

The mechanism

The local branch has no pretraining and cannot learn from 61 images; fusing a near-random branch into a strong pretrained one adds variance and costs accuracy — that contrast is negative on every seed. The Sobel prior is photometric, so it scores a lighting seam as highly as a defect; that is the mechanism we expected to hurt, and the point estimate agrees, but the signs disagree across seeds so it is not claimed on its own.

What survives

The negative result is about the architectural additions only. Untouched: the reproduction and audit of the base paper, the encrypted inference and its exactness bound, the dataset integrity audit — and the evaluation protocol, which is what made this visible instead of hiding it behind a saturated 100%.

What a follow-up should build

EfficientNet-B0 + the compression layer + the CKKS polynomial head — the privacy architecture without the extra feature machinery. It is simpler than what was proposed, it is what the evidence supports, and the encrypted inference results carry over unchanged because the head is identical.

Encrypted Inference · correctness

Does encryption change the prediction?

100.00%
encrypted vs plaintext agreement

1.43e-02
max absolute logit error

0
prediction flips out of 360

100.00%
accuracy under encryption

quantity	value
plaintext accuracy	100.00%
encrypted accuracy	100.00%
mean |logit error|	2.766e-03
test images evaluated	360

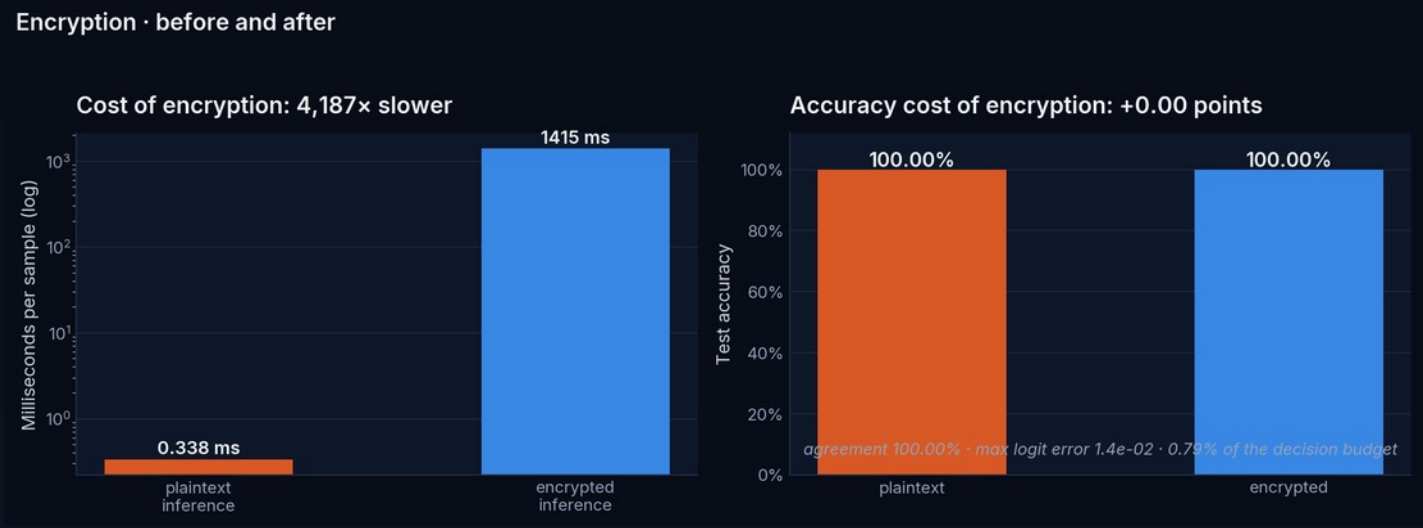
Why this is robust, not lucky

A prediction can only flip if the CKKS error exceeds half the gap between the top-1 and top-2 logits. On the tightest decision in the test set the error consumes just 0.79% of that budget — it would have to be ~126× larger before anything could change.

So agreement is a bound here, not a coincidence of this sample.

Encrypted Inference · cost

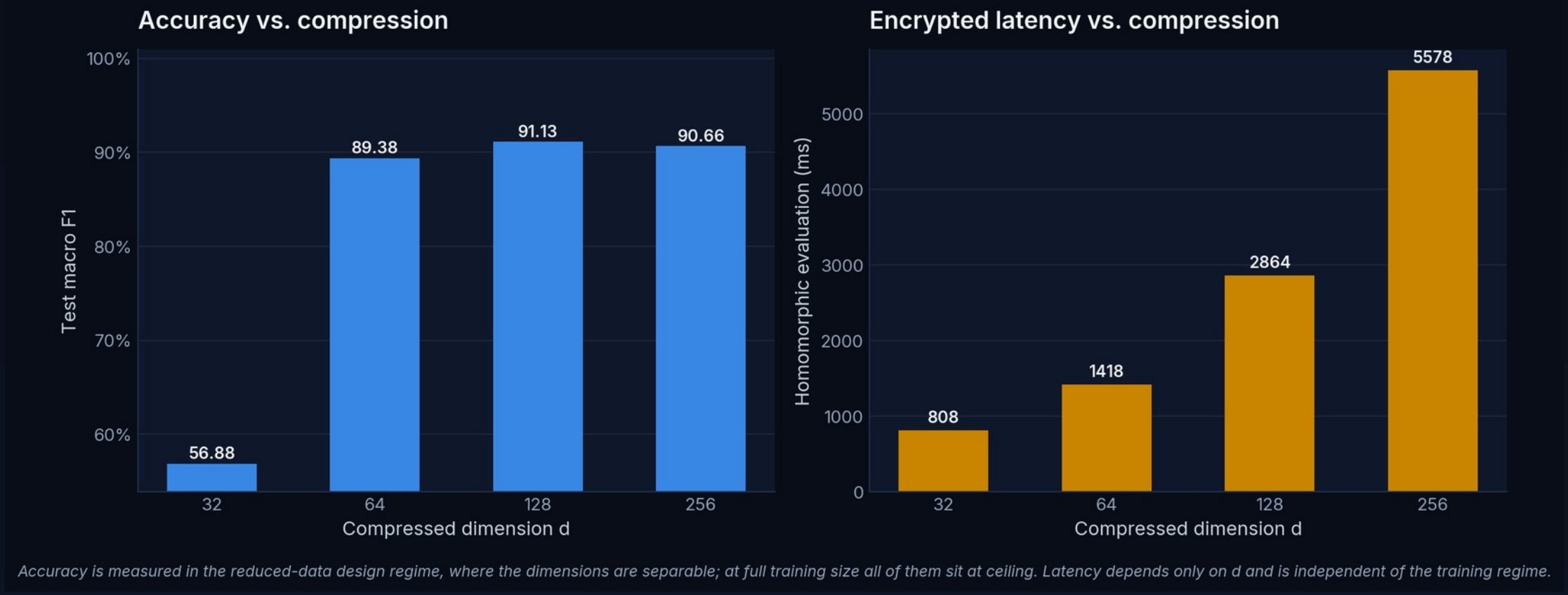
All timings on 4 vCPU, no GPU



quantity	value
client feature extraction	38 ms
encrypt (client)	16.8 ms
homomorphic evaluation	1415 ms
decrypt (client)	1.73 ms
ciphertext size	836 KB
expansion vs plaintext	3342×
server context (keys)	122 MB
peak RSS	1478 MB

Compression Trade-off

The compressed width d is the lever between accuracy and encrypted cost



Accuracy is measured in the reduced-data design regime, where the dimensions are separable; at full training size all of them sit at ceiling. Latency depends only on d and is independent of the training regime.

Accuracy here is measured in the reduced-data regime, where the dimensions are separable at all; at full training size they all sit at ceiling. Latency depends only on d and is independent of the training regime.

Threat Model · what is and is not protected

The comparison with the base paper is not like-for-like, and we do not present it as such

PROTECTED

— The feature vector in transit and at rest on the server.

— The prediction: the server returns ciphertext it cannot read.

— The secret key never leaves the client — asserted by test, not by assumption.

NOT PROTECTED

— The client must run and therefore hold the feature extractor — unlike the base paper, where the client does little work.

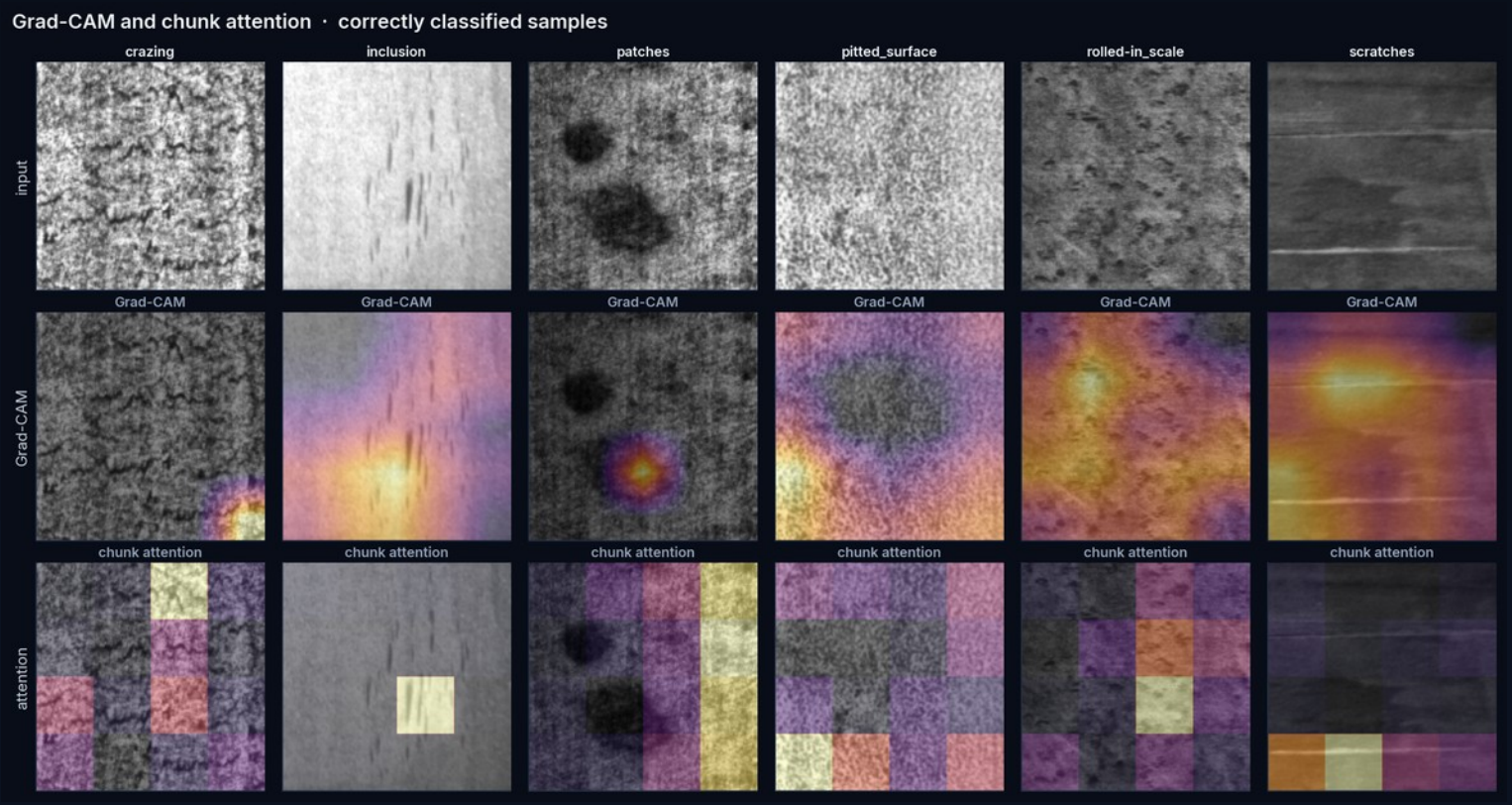
— Circuit privacy: an adaptive client could learn about the server's weights. Noise flooding is not implemented.

— The compressed vector is a lossy but real function of the image; we do not claim it is a formal privacy bound.

Honest summary

SAGE-HE protects the server side of the boundary. Jia et al. aim to protect the image itself. Ours is cheaper and exact; theirs is a stronger goal. These are different trade-offs, not a ranking.

Explainable AI · Grad-CAM and chunk attention

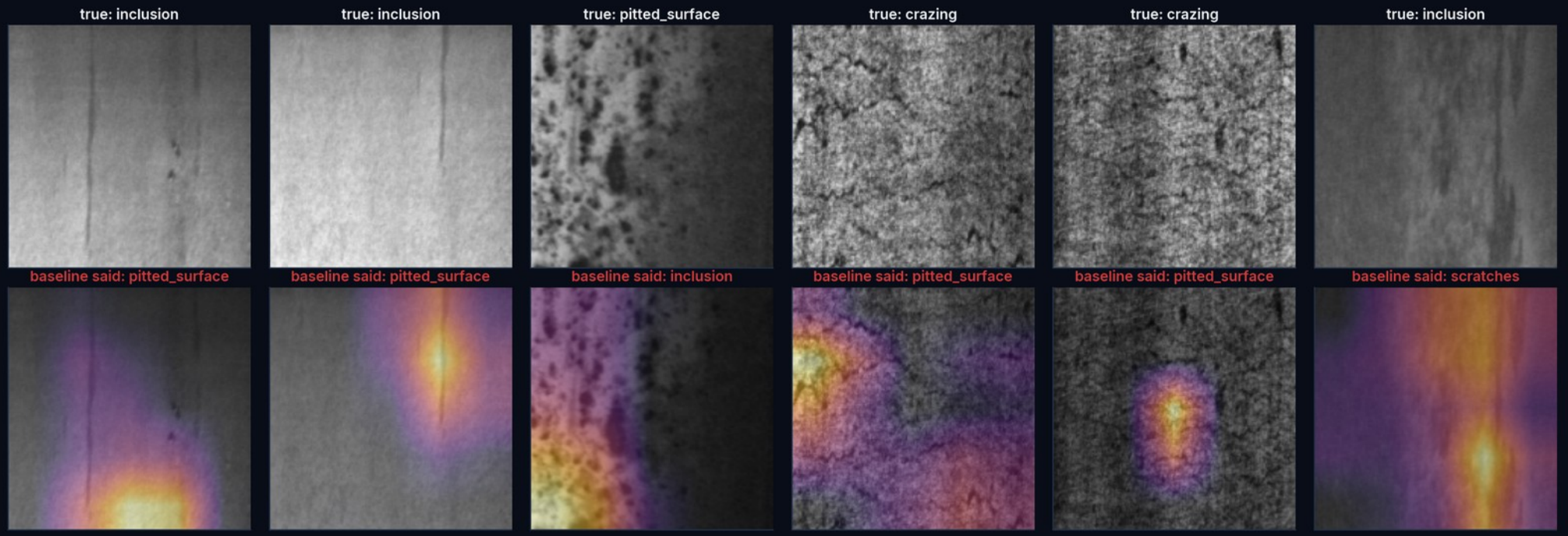


Grad-CAM is a post-hoc gradient attribution; chunk attention is the model's own selection weight. Agreement is weak evidence the model uses the defect region — an attractive heatmap is not proof of correctness.

Hard Cases

SAGE-HE misclassifies nothing on this test set, so these are the 19 images the reproduced baseline got wrong

Hard cases — the 19 images the reproduced baseline misclassified (SAGE-HE classifies all of them correctly)



The baseline's errors are concentrated in faint `inclusion` streaks it calls `pitted_surface` — the same two classes its per-class F1 singles out. These are shown instead of an empty failure slide; nothing here is curated.

Discussion and Limitations

Benchmark is saturated

NEU-CLS cannot separate strong modern architectures at full data. Conclusions about components come from the reduced-data regime.

Small test set

360 images. A one-image change moves accuracy by 0.28 points; Wilson intervals are reported for this reason.

No GPU

All training and all latencies are CPU. Timings are not comparable to GPU-accelerated figures elsewhere in the literature.

Weaker privacy than the base paper's goal

The client holds the extractor. We protect the server side only.

Baseline is a reconstruction

The paper under-specifies channel count, pooling, split and schedule; our reproduction documents each assumption.

Novelty is narrow

The encryption boundary is prior art. The contribution is the learned Sobel-conditioned selection and the measurement discipline.

Conclusion and Future Scope

— **Built.** A complete, tested, reproducible system: audited dataset, reproduced baseline, dual-branch model, genuine CKKS inference, explainability.

— **Measured.** SAGE-HE 100.00% vs reproduced baseline 93.98% on one identical held-out split, over 3 seeds.

— **Encrypted exactly.** Encrypted and plaintext inference agree, with a quantified margin: CKKS costs latency, not accuracy.

— **The architecture did not pan out.** Measured against its own backbone, the local branch, chunk attention and Sobel prior each cost accuracy. Reported, explained, and not claimed as a contribution.

FUTURE WORK

— A harder benchmark (NEU-DET, Severstal, X-SDD) where accuracy is not saturated.

— Higher-degree polynomial activations to recover the accuracy the square loses.

— Batched CKKS packing: 8192 slots are available and we use 64.

— Circuit privacy via noise flooding.

— Validate the λ finding on a dataset where gradient priors are less reliable.

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All entries resolved against Crossref or arXiv — see reports/citations_verified.json

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SAGE-HE

Thank you

RESEARCH, ARCHITECTURE & IMPLEMENTATION

SUHAIB S Z

Code, data manifests, logs and this deck are reproducible end to end